Chapter 5: Exponents

Step-by-Step Solutions to Exercise 5A

Generated by Gemini Notebook

September 2026

This document provides detailed, mathematically rigorous solutions to all problems in **Exercise 5A** of the chapter on **Exponents**. Every solution includes intermediate steps, clear derivations, and the specific Laws of Exponents applied.

Question 1: Evaluate

Evaluate each of the following expressions:

(i) 7^4

(ii) $(-5)^3$

(iii) $\left(\frac{3}{4}\right)^5$

(iv) $\left(\frac{-5}{2}\right)^2$

Solutions:

(i) **Evaluate 7^4:** By the definition of an exponent as repeated multiplication:

$$7^4 = 7 \times 7 \times 7 \times 7 = 49 \times 49 = 2401$$

Final Answer: 2401

(ii) **Evaluate $(-5)^3$:** Since the base is negative and the exponent is an odd positive integer, the result will be negative:

$$(-5)^3 = (-5) \times (-5) \times (-5) = 25 \times (-5) = -125$$

Final Answer: -125

(iii) **Evaluate $\left(\frac{3}{4}\right)^5$:** Using the law of exponents for a quotient, $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$:

$$\left(\frac{3}{4}\right)^5 = \frac{3^5}{4^5} = \frac{3 \times 3 \times 3 \times 3 \times 3}{4 \times 4 \times 4 \times 4 \times 4} = \frac{243}{1024}$$

Final Answer: $\frac{243}{1024}$

(iv) **Evaluate $\left(\frac{-5}{2}\right)^2$:** Using the law of exponents for a quotient, $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$:

$$\left(\frac{-5}{2}\right)^2 = \frac{(-5)^2}{2^2} = \frac{(-5) \times (-5)}{2 \times 2} = \frac{25}{4}$$

Final Answer: $\frac{25}{4}$

Question 2: Write as a power of 10

Write each of the following numbers as a power of 10:

(i) 10000

(ii) One crore

(iii) One million

Solutions:

(i) **Write 10000 as a power of 10:** Counting the number of zeros in 10000 (which is 4):

$$10000 = 10 \times 10 \times 10 \times 10 = 10^4$$

Final Answer: 10^4

(ii) **Write One crore as a power of 10:** In the Indian numbering system, one crore is equal to $1,00,00,000$. Counting the zeros (which is 7):

$$\text{One crore} = 10,000,000 = 10^7$$

Final Answer: 10^7

(iii) **Write One million as a power of 10:** In the international numbering system, one million is equal to $1,000,000$. Counting the zeros (which is 6):

$$\text{One million} = 1,000,000 = 10^6$$

Final Answer: 10^6

Question 3: Express in exponential notation

Express each of the following in exponential notation:

(i) $\left(\frac{-7}{13}\right) \times \left(\frac{-7}{13}\right) \times \left(\frac{-7}{13}\right)$

(ii) $\left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right)$

Solutions:

(i) Since the factor $\left(\frac{-7}{13}\right)$ is multiplied by itself 3 times, by definition of exponents:

$$\left(\frac{-7}{13}\right) \times \left(\frac{-7}{13}\right) \times \left(\frac{-7}{13}\right) = \left(\frac{-7}{13}\right)^3$$

Final Answer: $\left(\frac{-7}{13}\right)^3$

(ii) Since the factor $\left(\frac{-8}{3}\right)$ is multiplied by itself 4 times, by definition of exponents:

$$\left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right) \times \left(\frac{-8}{3}\right) = \left(\frac{-8}{3}\right)^4$$

Final Answer: $\left(\frac{-8}{3}\right)^4$

Question 4: Express in exponential notation

Express each of the following rational numbers in exponential notation:

(i) $\frac{343}{512}$

(ii) $\frac{-32}{243}$

(iii) $\frac{-1}{128}$

(iv) $\frac{729}{64}$

Solutions:

(i) **Express $\frac{343}{512}$:** We prime factorize the numerator and the denominator:

$$343 = 7 \times 7 \times 7 = 7^3$$
$$512 = 8 \times 8 \times 8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9 = 8^3$$

Using the quotient rule in reverse:

$$\frac{343}{512} = \frac{7^3}{8^3} = \left(\frac{7}{8}\right)^3$$

Final Answer: $\left(\frac{7}{8}\right)^3$

(ii) **Express $\frac{-32}{243}$:** Factorizing both:

$$-32 = (-2) \times (-2) \times (-2) \times (-2) \times (-2) = (-2)^5$$
$$243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^5$$

Thus:

$$\frac{-32}{243} = \frac{(-2)^5}{3^5} = \left(\frac{-2}{3}\right)^5$$

Final Answer: $\left(\frac{-2}{3}\right)^5$

(iii) **Express $\frac{-1}{128}$:** Factorizing:

$$-1 = (-1)^7 \quad (\text{since } (-1)^{\text{odd}} = -1)$$
$$128 = 2^7$$

Thus:

$$\frac{-1}{128} = \frac{(-1)^7}{2^7} = \left(\frac{-1}{2}\right)^7$$

Final Answer: $\left(\frac{-1}{2}\right)^7$

(iv) **Express $\frac{729}{64}$:** We can write this in two different exponential forms:

- **Form A (base 3/2):**

$$729 = 3^6$$
$$64 = 2^6$$

  So, $\frac{729}{64} = \frac{3^6}{2^6} = \left(\frac{3}{2}\right)^6$.

- **Form B (base 9/4):**

$$729 = 9^3$$
$$64 = 4^3$$

  So, $\frac{729}{64} = \frac{9^3}{4^3} = \left(\frac{9}{4}\right)^3$.

Both are perfectly valid and correct.

Final Answer: $\left(\frac{3}{2}\right)^6$ **or** $\left(\frac{9}{4}\right)^3$

Question 5: Express in exponential notation

Using the laws of exponents, simplify and express each of the following in exponential notation:

(i) $\left(\frac{5}{21}\right)^3 \times \left(\frac{5}{21}\right)^8$

(ii) $\left(\frac{-7}{3}\right)^{11} \times \left(\frac{-7}{3}\right)^{13}$

(iii) $\left(\frac{13}{43}\right)^7 \div \left(\frac{13}{43}\right)^2$

(iv) $\left(\frac{-16}{35}\right)^{16} \div \left(\frac{-16}{35}\right)^3$

(v) $\left(\frac{-7}{15}\right)^{12} \div \left(\frac{-7}{15}\right)^{15}$

(vi) $\left(\frac{1}{24}\right)^{24} \div \left(\frac{1}{24}\right)^{16}$

Solutions:

Laws of Exponents Applied

We use the following core rules:

1. **Product Law:** $a^m \times a^n = a^{m+n}$
2. **Quotient Law:** $a^m \div a^n = a^{m-n}$ (or $\frac{1}{a^{n-m}}$ if $m < n$)

(i) **Simplify $\left(\frac{5}{21}\right)^3 \times \left(\frac{5}{21}\right)^8$:** Applying the Product Law with base $a = \frac{5}{21}$:

$$\left(\frac{5}{21}\right)^3 \times \left(\frac{5}{21}\right)^8 = \left(\frac{5}{21}\right)^{3+8} = \left(\frac{5}{21}\right)^{11}$$

Final Answer: $\left(\frac{5}{21}\right)^{11}$

(ii) **Simplify $\left(\frac{-7}{3}\right)^{11} \times \left(\frac{-7}{3}\right)^{13}$:** Applying the Product Law with base $a = \frac{-7}{3}$:

$$\left(\frac{-7}{3}\right)^{11} \times \left(\frac{-7}{3}\right)^{13} = \left(\frac{-7}{3}\right)^{11+13} = \left(\frac{-7}{3}\right)^{24}$$

Final Answer: $\left(\frac{-7}{3}\right)^{24}$

(iii) **Simplify $\left(\frac{13}{43}\right)^7 \div \left(\frac{13}{43}\right)^2$:** Applying the Quotient Law with base $a = \frac{13}{43}$ since $7 > 2$:

$$\left(\frac{13}{43}\right)^7 \div \left(\frac{13}{43}\right)^2 = \left(\frac{13}{43}\right)^{7-2} = \left(\frac{13}{43}\right)^5$$

Final Answer: $\left(\frac{13}{43}\right)^5$

(iv) **Simplify** $\left(\frac{-16}{35}\right)^{16} \div \left(\frac{-16}{35}\right)^{3}$**:** Applying the Quotient Law with base $a = \frac{-16}{35}$ since $16 > 3$:

$$\left(\frac{-16}{35}\right)^{16} \div \left(\frac{-16}{35}\right)^{3} = \left(\frac{-16}{35}\right)^{16-3} = \left(\frac{-16}{35}\right)^{13}$$

Final Answer: $\left(\frac{-16}{35}\right)^{13}$

(v) **Simplify** $\left(\frac{-7}{15}\right)^{12} \div \left(\frac{-7}{15}\right)^{15}$**:** Applying the Quotient Law where $m < n$:

$$\left(\frac{-7}{15}\right)^{12} \div \left(\frac{-7}{15}\right)^{15} = \left(\frac{-7}{15}\right)^{12-15} = \left(\frac{-7}{15}\right)^{-3}$$

To express this with a positive exponent, we take the reciprocal of the base:

$$\left(\frac{-7}{15}\right)^{-3} = \left(\frac{15}{-7}\right)^{3} = \left(\frac{-15}{7}\right)^{3} = \frac{1}{\left(\frac{-7}{15}\right)^{3}}$$

Final Answer: $\left(\frac{-15}{7}\right)^{3}$

(vi) **Simplify** $\left(\frac{1}{24}\right)^{24} \div \left(\frac{1}{24}\right)^{16}$**:** Applying the Quotient Law with base $a = \frac{1}{24}$ since $24 > 16$:

$$\left(\frac{1}{24}\right)^{24} \div \left(\frac{1}{24}\right)^{16} = \left(\frac{1}{24}\right)^{24-16} = \left(\frac{1}{24}\right)^{8}$$

Final Answer: $\left(\frac{1}{24}\right)^{8}$

Question 6: Simplify and express as a rational number

Simplify each of the following expressions and express the result as a rational number (in the form $\frac{p}{q}$):

(i) $\left(\frac{6}{5}\right)^3 \times \left(\frac{5}{2}\right)^2$

(ii) $\left(\frac{3}{4}\right)^2 \times \left(\frac{-1}{2}\right)^5 \times 2^3$

(iii) $\left(\frac{5}{4}\right)^2 \times \left(\frac{2}{3}\right)^3 \times \left(\frac{-3}{5}\right)^3$

(iv) $\left(\frac{-3}{4}\right)^3 \times \left(\frac{-5}{2}\right)^3 \times \left(\frac{2}{3}\right)^5$

(v) $\left\{\left(\frac{7}{11}\right)^6 \div \left(\frac{7}{11}\right)^3\right\}^3$

(vi) $\left(\frac{-4}{3}\right)^8 \div \left(\frac{-4}{3}\right)^{12}$

Solutions:

(i) **Simplify $\left(\frac{6}{5}\right)^3 \times \left(\frac{5}{2}\right)^2$:** Using the power rules:

$$\left(\frac{6}{5}\right)^3 \times \left(\frac{5}{2}\right)^2 = \frac{6^3}{5^3} \times \frac{5^2}{2^2} = \frac{216}{125} \times \frac{25}{4} = \frac{216}{4} \times \frac{25}{125} = 54 \times \frac{1}{5} = \frac{54}{5}$$

Final Answer: $\frac{54}{5}$

(ii) **Simplify $\left(\frac{3}{4}\right)^2 \times \left(\frac{-1}{2}\right)^5 \times 2^3$:**

$$\left(\frac{3}{4}\right)^2 \times \left(\frac{-1}{2}\right)^5 \times 2^3 = \frac{9}{16} \times \left(\frac{-1}{32}\right) \times 8$$

$$= \frac{9 \times (-1) \times 8}{16 \times 32} = \frac{-72}{512} = -\frac{9}{64}$$

Final Answer: $-\frac{9}{64}$

(iii) **Simplify $\left(\frac{5}{4}\right)^2 \times \left(\frac{2}{3}\right)^3 \times \left(\frac{-3}{5}\right)^3$:** Expanding the exponents:

$$\frac{5^2}{4^2} \times \frac{2^3}{3^3} \times \frac{(-3)^3}{5^3} = \frac{25}{16} \times \frac{8}{27} \times \left(-\frac{27}{125}\right)$$

$$= \left(\frac{25}{125}\right) \times \left(\frac{8}{16}\right) \times \left(\frac{-27}{27}\right)$$

$$= \frac{1}{5} \times \frac{1}{2} \times (-1) = -\frac{1}{10}$$

Final Answer: $-\frac{1}{10}$

(iv) **Simplify** $\left(\frac{-3}{4}\right)^3 \times \left(\frac{-5}{2}\right)^3 \times \left(\frac{2}{3}\right)^5$**:** Expanding the terms:

$$\frac{(-3)^3}{4^3} \times \frac{(-5)^3}{2^3} \times \frac{2^5}{3^5} = \frac{-27}{64} \times \frac{-125}{8} \times \frac{32}{243}$$
$$= \frac{(-27) \times (-125) \times 32}{64 \times 8 \times 243}$$

Let's simplify: - $\frac{32}{64} = \frac{1}{2}$ - $\frac{-27}{243} = \frac{-1}{9}$ Substituting these in:

$$\frac{(-1) \times (-125) \times 1}{2 \times 8 \times 9} = \frac{125}{144}$$

Final Answer: $\frac{125}{144}$

(v) **Simplify** $\left\{\left(\frac{7}{11}\right)^6 \div \left(\frac{7}{11}\right)^3\right\}^3$**:** First, simplify inside the curly braces using the Quotient Law $a^m \div a^n = a^{m-n}$:

$$\left(\frac{7}{11}\right)^6 \div \left(\frac{7}{11}\right)^3 = \left(\frac{7}{11}\right)^{6-3} = \left(\frac{7}{11}\right)^3$$

Now apply the outer exponent using the Power of a Power rule $(a^m)^n = a^{mn}$:

$$\left\{\left(\frac{7}{11}\right)^3\right\}^3 = \left(\frac{7}{11}\right)^{3\times3} = \left(\frac{7}{11}\right)^9$$

Evaluating this as a rational number:

$$7^9 = 40,353,607, \quad 11^9 = 2,357,947,691 \implies \frac{40,353,607}{2,357,947,691}$$

Leaving it in the simplified exponential form is preferred for such large numbers, but we provide both here.

Final Answer: $\left(\frac{7}{11}\right)^9 = \frac{40{,}353{,}607}{2{,}357{,}947{,}691}$

(vi) **Simplify** $\left(\frac{-4}{3}\right)^8 \div \left(\frac{-4}{3}\right)^{12}$**:** Applying the Quotient Law with base $a = \frac{-4}{3}$ and $m < n$:

$$\left(\frac{-4}{3}\right)^8 \div \left(\frac{-4}{3}\right)^{12} = \left(\frac{-4}{3}\right)^{8-12} = \left(\frac{-4}{3}\right)^{-4}$$

Convert to a positive exponent by taking the reciprocal of the base:

$$\left(\frac{-4}{3}\right)^{-4} = \left(\frac{3}{-4}\right)^4 = \frac{3^4}{(-4)^4} = \frac{81}{256}$$

Final Answer: $\frac{81}{256}$

Question 7: Simplify

Simplify the following expressions and write as a rational number:

(i) $\frac{10^2\times15^3}{2^2\times3\times5^5\times6^4}$

(ii) $\frac{3^5\times25\times10^5}{5^7\times6^5}$

Solutions:

(i) **Simplify** $\frac{10^2\times15^3}{2^2\times3\times5^5\times6^4}$**:** We express each base as a product of its prime factors: - $10 = 2\times5 \implies 10^2 = (2\times5)^2 = 2^2\times5^2$ - $15 = 3\times5 \implies 15^3 = (3\times5)^3 = 3^3\times5^3$ - $6 = 2\times3 \implies 6^4 = (2\times3)^4 = 2^4\times3^4$

Substitute these prime factorized forms back into the expression:

$$\text{Numerator} = 10^2\times15^3 = (2^2\times5^2)\times(3^3\times5^3) = 2^2\times3^3\times5^{2+3} = 2^2\times3^3\times5^5$$

$$\text{Denominator} = 2^2\times3\times5^5\times6^4 = 2^2\times3^1\times5^5\times(2^4\times3^4) = 2^{2+4}\times3^{1+4}\times5^5 = 2^6\times3^5\times5^5$$

Now, divide the numerator by the denominator:

$$\frac{2^2\times3^3\times5^5}{2^6\times3^5\times5^5} = \frac{2^2}{2^6}\times\frac{3^3}{3^5}\times\frac{5^5}{5^5} = 2^{2-6}\times3^{3-5}\times1 = 2^{-4}\times3^{-2}$$

Convert to positive exponents:

$$\frac{1}{2^4\times3^2} = \frac{1}{16\times9} = \frac{1}{144}$$

Final Answer: $\frac{1}{144}$

(ii) **Simplify** $\frac{3^5\times25\times10^5}{5^7\times6^5}$**:** Express in prime factorized form: - $25 = 5^2$ - $10^5 = (2\times5)^5 = 2^5\times5^5$ - $6^5 = (2\times3)^5 = 2^5\times3^5$

Substitute back:

$$\text{Numerator} = 3^5\times5^2\times(2^5\times5^5) = 2^5\times3^5\times5^{2+5} = 2^5\times3^5\times5^7$$

$$\text{Denominator} = 5^7\times(2^5\times3^5) = 2^5\times3^5\times5^7$$

Notice that the numerator and the denominator are exactly identical:

$$\frac{2^5\times3^5\times5^7}{2^5\times3^5\times5^7} = 1$$

Final Answer: 1

Question 8: Distance Earth to Moon

The distance between the Earth and the Moon is approximately $384,000$ km. Express this distance in metres in exponential notation.

Solution:

1. First, we recall the relation between kilometres (km) and metres (m):

$$1 \text{ km} = 1000 \text{ m} = 10^3 \text{ m}$$

2. Convert the given distance of $384,000$ km into metres:

$$384,000 \text{ km} = 384,000 \times 10^3 \text{ m} = 384,000,000 \text{ m}$$

3. Express $384,000,000$ m in exponential notations:

   - **In standard scientific notation:**

$$3.84 \times 10^8 \text{ m}$$

   - **Following the textbook's visual pattern of direct conversion ($d \times 10^3$):**

$$384 \times 10^6 \text{ m} \quad \text{or} \quad 38,4000 \times 10^3 \text{ m}$$

Final Answer: 3.84×10^8 **m**

Question 9: RAM of Computer

The RAM of a computer is 8 gigabyte. If each gigabyte is equal to 10^9 bytes, then express the RAM in bytes.

Solution:

1. Given RAM size $= 8$ GB.

2. Since 1 gigabyte (GB) $= 10^9$ bytes:

$$8 \text{ gigabytes} = 8 \times 10^9 \text{ bytes}$$

Final Answer: 8×10^9 **bytes**

Question 10: Tennis Competition

In a tennis competition, 128 players were selected for a series of knockout rounds. In each round the losers were eliminated and the winners reached the next round. How many players moved to the next round after the 4th round? Express this number in exponential notation in terms of the initial number of players.

Solution:

1. **Determine the number of remaining players step-by-step:** In a knockout tournament, each match halves the number of players.
   - **Initial players:** 128 players.
   - **After Round 1:** $\frac{128}{2} = 64$ players remain.
   - **After Round 2:** $\frac{64}{2} = 32$ players remain.
   - **After Round 3:** $\frac{32}{2} = 16$ players remain.
   - **After Round 4:** $\frac{16}{2} = 8$ players remain.

   So, 8 players moved to the next round after the 4th round.
2. **Express this in terms of the initial number of players (128):** Since the number of players halves after each round, after 4 rounds, the initial number of players is divided by 2^4:

$$\text{Players remaining} = \frac{128}{2^4} = 128 \times 2^{-4}$$

3. **Express both initial and final players in powers of 2:**

$$\text{Initial players } (128) = 2^7$$
$$\text{Final players } (8) = 2^3 = \frac{2^7}{2^4} = 2^{7-4}$$

Final Answer: 8 **players** $= 2^3$ **players** $= \frac{128}{2^4}$ **players**

Question 11: Unit Conversion

Express the following in centimetres (cm) in exponential notation:

(i) 98 hm

(ii) 156 km

(iii) 371 m

Solution: First, we establish the conversion metrics from the text's "Units of Length" section:

- $1 \text{ m} = 10 \text{ dm} = 10 \times 10 \text{ cm} = 100 \text{ cm} = 10^2 \text{ cm}$
- $1 \text{ hm} = 10 \text{ dam} = 10 \times 10 \text{ m} = 100 \text{ m} = 100 \times 100 \text{ cm} = 10,000 \text{ cm} = 10^4 \text{ cm}$
- $1 \text{ km} = 1000 \text{ m} = 1000 \times 100 \text{ cm} = 100,000 \text{ cm} = 10^5 \text{ cm}$

Let's apply these conversions:

(i) **Convert 98 hm to cm:**

$$98 \text{ hm} = 98 \times 10^4 \text{ cm} = 9.8 \times 10^5 \text{ cm}$$

Final Answer: 9.8×10^5 **cm**

(ii) **Convert 156 km to cm:**

$$156 \text{ km} = 156 \times 10^5 \text{ cm} = 1.56 \times 10^7 \text{ cm}$$

Final Answer: 1.56×10^7 **cm**

(iii) **Convert 371 m to cm:**

$$371 \text{ m} = 371 \times 10^2 \text{ cm} = 3.71 \times 10^4 \text{ cm}$$

Final Answer: 3.71×10^4 **cm**